

FITNESS CENTER MANAGEMENT PLATFORM

Complete Product Specification

Odoo 19 | Frontend-First Architecture

Comprehensive Research: Nike NTC | Peloton | Freeletics | Strava | cult.fit | Down Dog |
Fitbit

6 Product Verticals | 35+ Models | 8 OWL Components | 25+ Feature Categories
Frontend-First Design | AI Engine | Social & Gamification | Wellness Ecosystem

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1. Executive Summary

This document defines the complete product specification for a world-class Fitness Center Management Platform built on Odoo 19. Unlike the previous sprint-focused plan, this specification prioritizes building the best possible product by drawing inspiration from seven industry leaders.

Our platform uniquely combines gym operations management (like Mindbody/GymMaster) with personal fitness tracking (like Fitbit), social engagement (like Strava), AI-powered coaching (like Freeletics), guided wellness (like cult.fit Mind.fit + Down Dog), nutrition management (like cult.fit Eat.fit), and premium class experiences (like Peloton) into a single integrated Odoo module.

1.1 Product Vision

Build a cult.fit-inspired fitness ecosystem within Odoo 19 that serves as the single platform for gym owners, trainers, and members, covering the complete fitness journey from sign-up to transformation.

1.2 Key Differentiators

- All-in-One Ecosystem: Six product verticals (FIT, WORKOUT, EAT, MIND, SOCIAL, CARE) in one Odoo module, unlike competitors that only cover 1-2 areas.
- Frontend-First Design: OWL 3.x components drive the experience, with 8 custom widgets (dashboard, workout timer, activity feed, check-in kiosk, leaderboard, meal tracker, meditation player, body scanner).
- AI Coach: Rule-based recommendation engine (ML-ready) that personalizes workouts, meals, and wellness sessions based on member profile, history, and goals. Inspired by Freeletics AI Coach.
- Gym + Personal Fitness: Manages gym operations (equipment, classes, trainers, billing) AND personal fitness tracking (workouts, nutrition, body metrics, sleep), unlike any single competitor.
- Social-First: Strava-inspired social network with friends, activity feed, kudos, leaderboards, challenges, achievements, and clubs, making fitness a community experience.
- cult.fit Wellness: Mind.fit-inspired meditation, sleep tracking, stress management, and Down Dog-style AI-generated yoga sessions with customizable pace/focus/duration.

2. Competitor Research & Analysis

We analyzed seven leading platforms across fitness, wellness, and gym management to extract the best features from each:

Platform	Key Features	What We Take
Nike Training Club	185+ free workouts, trainer-led video sessions, 4-6 week adaptive plans, Apple Watch integration, NTC TV with recipes + meditation, badge/trophy gamification, workout scheduling + casting to TV, yoga + pilates + HIIT + strength.	Free content model, video quality, adaptive plans, Apple ecosystem integration
Peloton	Live + on-demand classes, real-time leaderboard during workouts, High Five social feature, Teams (50K members), Club Peloton loyalty tiers, Strive Score (heart rate zones), target metrics for cadence/resistance/speed, instructor-led community.	Live class experience, real-time competition, loyalty program, instructor personality
Freeletics	AI Coach with 56M+ user data, 700+ exercises, 1 trillion workout combinations, Coach+ conversational AI, real-time rep counting from camera, nutrition coaching with meal plans, mindset audio coaching, bodyweight + gym + running.	AI personalization depth, no-equipment workouts, nutrition integration, mindset coaching
Strava	GPS activity tracking, segments with KOM/QOM records, 1M+ clubs, group challenges, filtered leaderboards (age/weight/club), social feed with kudos/comments, race search, Year in Sport analytics, 14B kudos in 2025.	Social network effect, competitive segments, club community, data-rich analytics
cult.fit	4-vertical ecosystem (Fit/Eat/Mind/Care), group classes (boxing/yoga/HRX), cultpass tiers (LIVE/PRO/ELITE), Eat.fit meal delivery, Mind.fit meditation, Care.fit health consultations, energy meter gamification, body composition analysis.	Ecosystem approach, physical + digital integration, health verticals, meal delivery
Down Dog	AI-generated unique sessions every time, customizable style/pace/focus/music/voice, yoga + HIIT + barre + meditation + prenatal, difficulty progression, offline download, no two workouts alike.	AI session generation, extreme customization, multi-discipline, personalization depth
Fitbit	24/7 heart rate, sleep stages + Sleep Score + Sleep Profile, stress management score (cEDA sensor), SpO2, glucose tracking, Gemini AI health coach, Daily Readiness Score, Active Zone Minutes, 40+ exercise modes.	Wearable data richness, health metrics depth, sleep analysis, AI health coach

The comprehensive feature matrix below shows how our platform covers all 25 feature categories, while no single competitor covers more than 15:

COMPREHENSIVE COMPETITOR FEATURE MATRIX

Feature	Nike NTC	Peloton	Freeletics	Strava	cult.fit	Down Dog	Fitbit
WORKOUT FEATURES							
Workout Plans							
Video Coaching							
Workout Timer							
AI Coach							
Custom Builder							
NUTRITION FEATURES							
Food Database							
Meal Planning							
Calorie Tracking							
SOCIAL FEATURES							
Friends							
Leaderboards							
Challenges							
TRACKING FEATURES							
Heart Rate Track							
Sleep Tracking							

OUR MODULE: 25/25 Features

Workout Features							
Workout Plans							
Video Coaching							
Workout Timer							
AI Coach							
Custom Builder							
Nutrition Features							
Food Database							
Meal Planning							
Calorie Tracking							
Social Features							
Friends							
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OUR MODULE: 25/25 Features

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TRACKING FEATURES							
Heart Rate Tracking							
Sleep Tracking							

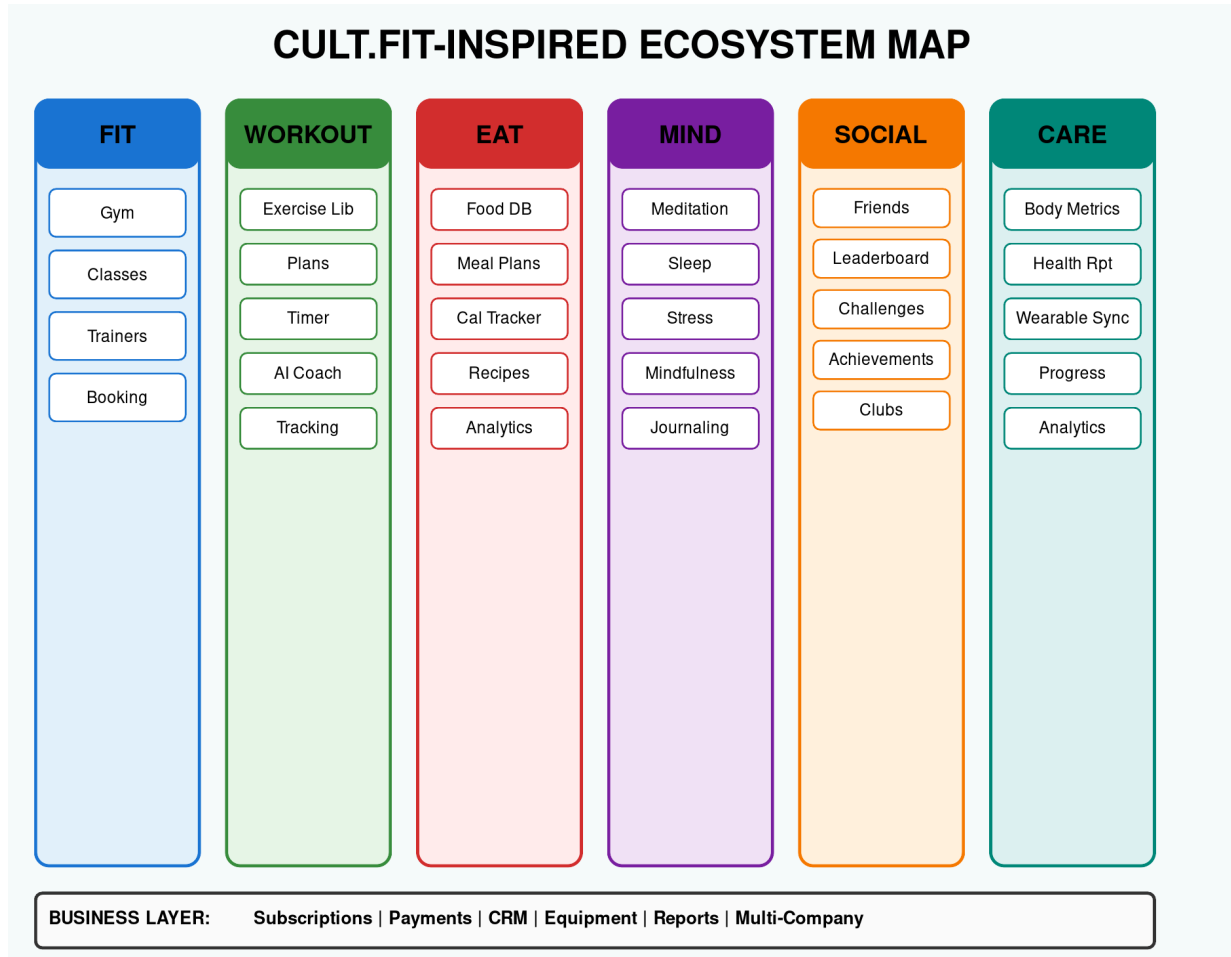
TRACKING FEATURES								
Heart Rate Tracking								
Sleep Tracking								

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Heart Rate Trac								
Sleep Tracking								

OUR MODULE: 25/25 Features

3. Product Ecosystem (cult.fit-Inspired)

Inspired by cult.fit's four-vertical model, we expand to six product verticals plus a comprehensive business layer, all integrated within a single Odoo 19 module:



3.1 Vertical Overview

FIT (Gym + Classes)

Gym access management, group fitness classes (yoga, boxing, HIIT, dance, strength), class scheduling with calendar views, trainer profiles with booking, room/facility management, capacity control. Inspired by: cult.fit group classes + Peloton instructor-led sessions.

WORKOUT (Exercise Engine)

700+ exercise library with video demos, custom workout builder, predefined plans (4-12 weeks), real-time session tracker with OWL timer (HIIT/Tabata/AMRAP/EMOM), workout logging with PR tracking, progressive overload charts. Inspired by: Nike NTC plans + Freeletics AI + Down Dog customization.

EAT (Diet & Nutrition)

Food database (USDA preloaded + custom), daily meal logging with quantities, calorie tracker (target vs consumed vs burned), meal planner (weekly, goal-based), macro breakdown (protein/carbs/fats), recipe suggestions, water intake. Inspired by: cult.fit Eat.fit + Freeletics nutrition.

MIND (Wellness)

Guided meditation sessions with timer and ambient music, sleep tracking with quality score, stress management with daily score and breathing exercises, AI-generated yoga sessions (customizable style/pace/focus like Down Dog), mindfulness journals. Inspired by: cult.fit Mind.fit + Down Dog AI sessions.

SOCIAL (Community)

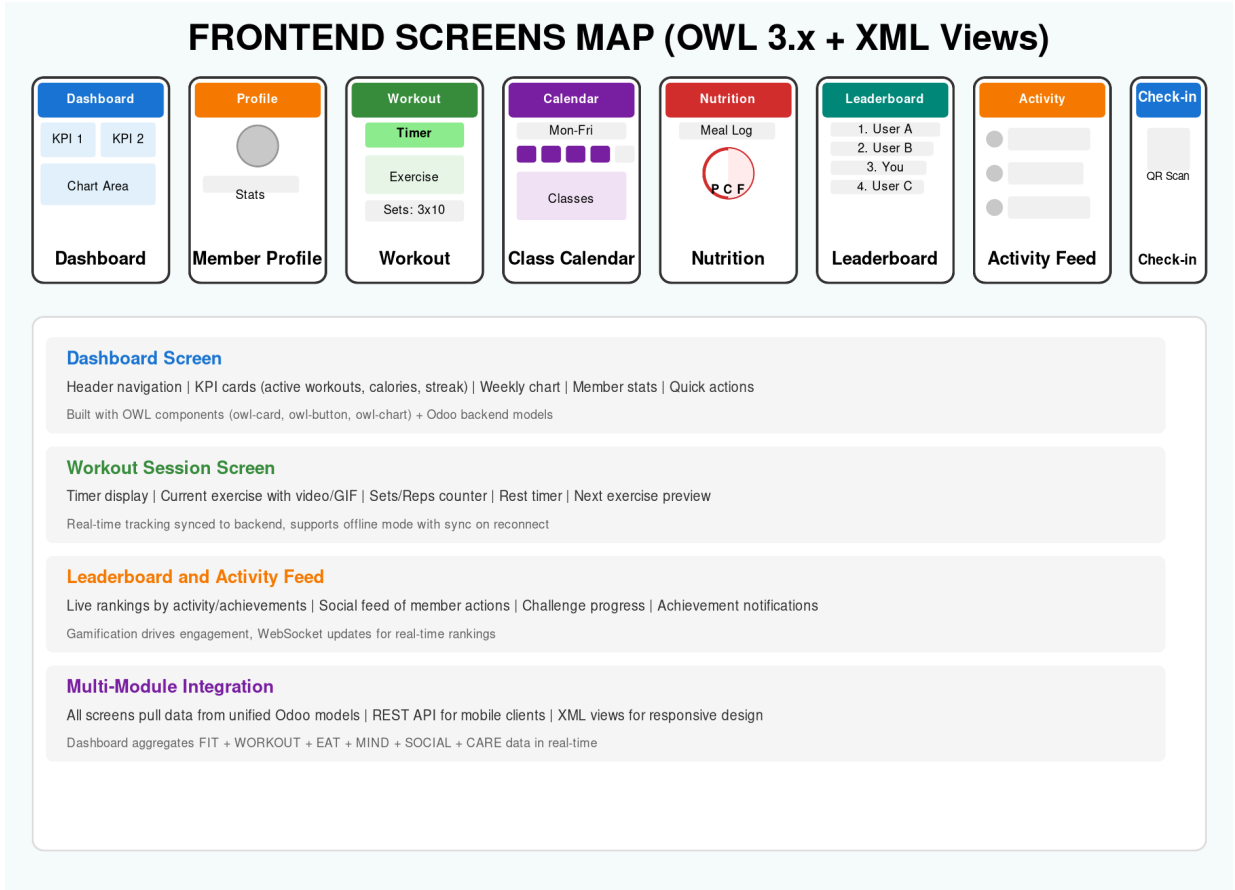
Friends/followers with activity feed, kudos and comments, weekly/monthly leaderboards (filterable), group challenges (distance/workout count/streak), achievement badges (bronze through platinum), clubs with shared leaderboards and events. Inspired by: Strava social + Peloton Teams + cult.fit energy meter.

CARE (Health Tracking)

Body metrics (weight/BMI/body fat/measurements with trend graphs), progress snapshots with before/after photos, weekly/monthly progress reports, wearable-ready data sync (API hooks for Fitbit/Google Fit/Apple Health), Daily Readiness Score, sleep quality integration. Inspired by: Fitbit health metrics + Strava analytics.

4. Frontend Screens & UX Design

The platform follows a frontend-first approach: we design the user experience first, then build the backend to support it. All interactive components use OWL 3.x, while standard CRUD uses Odoo XML views.



4.1 OWL 3.x Custom Components (8 Widgets)

Component	Description
Dashboard Widget	Client action with 6 KPI cards + 4 chart panels. Real-time data via RPC. Responsive grid layout. Charts via Chart.js. Panels: membership trend, revenue by plan, attendance heatmap, workout popularity, nutrition compliance, social pulse.
Workout Timer Widget	Full-screen OWL component for active workout sessions. Modes: countdown, interval (HIIT), Tabata (20/10), AMRAP, EMOM. Visual: large timer, exercise name, set/rep counter, rest alert animation, calorie burn counter, progress bar.
Activity Feed Widget	Strava-inspired social feed showing friend workouts, achievements, challenge progress. Supports kudos (tap to high-five), comments, and share. Infinite scroll with lazy loading.
Check-in Kiosk Widget	Tablet-optimized full-screen component for gym entrance. Member search by name/barcode/QR. Photo display on match for verification. Large check-in/check-out toggle. Success/error animations. Works offline with sync.

Leaderboard Widget	Real-time rankings with filters (period, metric, age group, club). Animated rank changes. Current member highlighted. Supports: workout count, calories, streaks, volume lifted. Tabs for weekly/monthly/all-time.
Meal Tracker Widget	Quick food logging with search-as-you-type. Daily view with breakfast/lunch/dinner/snack sections. Macro donut chart (protein/carbs/fats). Calorie progress bar (consumed vs target). Recent meals for quick re-log.
Meditation Player Widget	Guided session player with ambient background sounds. Timer modes: guided (with audio cues), silent, breathing (inhale/hold/exhale animation). Session categories: focus, sleep, stress relief, morning, evening.
Body Scanner Widget	Visual body measurement input with human silhouette. Tap body parts to enter measurements. Before/after photo comparison slider. Trend charts for each metric. BMI/body fat calculators with visual indicators.

4.2 Standard Odoo XML Views

Beyond OWL components, all models get standard Odoo views: form views with statusbar for stateful models, list views with activity widgets, kanban views with photo cards (members, trainers), calendar views for schedules and classes, search views with smart filters and groupby, pivot + graph views for analytics. All views use v19 syntax with direct invisible/readonly Python expressions.

5. FIT Module: Gym, Classes & Trainers

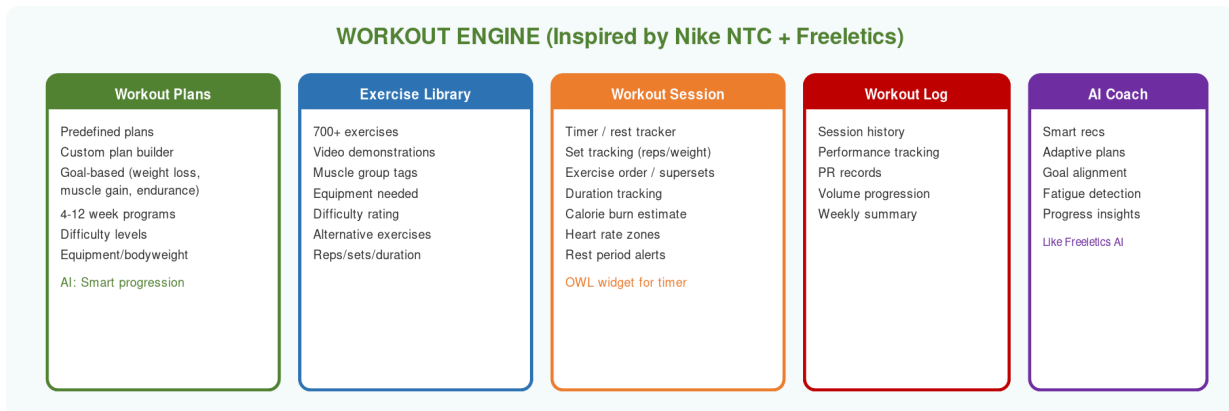
The FIT module manages physical gym operations: member check-in/check-out, group fitness class scheduling, trainer management and personal training bookings, room allocation, and capacity management.

5.1 Models

Model	Key Fields
fitness.class	name, class_type (Selection: yoga/boxing/HIIT/dance/strength/pilates/CrossFit/bootcamp), description (Html), default_duration, max_capacity, difficulty_level, equipment_needed, calories_estimate, video_preview_url, active
fitness.class.schedule	class_id (M2O), trainer_id (M2O), date_start (Datetime), date_end (Datetime), room (Char), current_bookings (Computed), available_spots (Computed), state (scheduled/in_progress/completed/cancelled), recurring (Boolean), recurrence_rule
fitness.class.booking	schedule_id (M2O), member_id (M2O), booking_date, state (booked/waitlisted/attended/cancelled/no_show), waitlist_position, auto_promote (Boolean)
fitness.trainer	name, employee_id (M2O hr.employee), specialization_ids (M2M), certification, experience_years, hourly_rate, bio (Html), photo (Binary), rating (Computed avg), availability_schedule, max_clients, active
fitness.trainer.booking	trainer_id (M2O), member_id (M2O), datetime, duration, type (Selection: personal/group/assessment/consultation), state, notes, price
fitness.room	name, capacity, equipment_available, floor, active
fitness.attendance	member_id (M2O), check_in (Datetime), check_out, duration (Computed), method (manual/qr/barcode/rfid), facility_id

6. WORKOUT Module: Exercise Engine & AI Coach

Inspired by Nike NTC's curated plans, Freeletics's AI coach, and Down Dog's AI-generated sessions, this module provides a comprehensive workout engine with 700+ exercises, custom plan builder, real-time session tracking, and AI-powered recommendations.

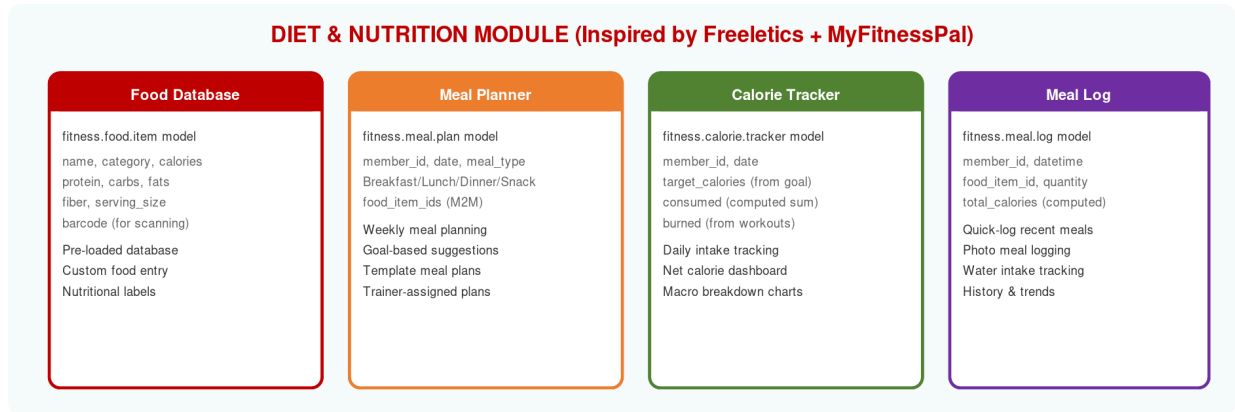


6.1 Models

Model	Key Fields
fitness.workout.plan	name, goal (Selection: weight_loss/muscle_gain/endurance/flexibility/general), difficulty (beginner/intermediate/advanced), duration_weeks, sessions_per_week, exercise_ids (M2M through fitness.workout.plan.line), description, creator_id (trainer or system), is_template, is_ai_generated
fitness.workout.plan.line	plan_id, exercise_id, day_number, order, sets, reps, duration_seconds, rest_seconds, notes
fitness.workout.exercise	name, category_id (M2O), video_url, thumbnail (Binary), muscle_groups (M2M), equipment_ids (M2M), difficulty, calories_per_minute, instructions (Html), alternatives_ids (M2M self), is_bodyweight
fitness.exercise.category	name (cardio/strength/flexibility/HIIT/yoga/pilates/boxing/dance), description, icon_color
fitness.workout.session	member_id, plan_id, start_time, end_time, status (active/paused/completed/abandoned), total_duration, calories_burned, intensity_score, heart_rate_avg, notes, mood_before, mood_after
fitness.workout.log	session_id, exercise_id, set_number, reps_done, weight_used, duration_seconds, is_pr (Boolean), rest_taken
fitness.workout.timer	session_id, mode (countdown/interval/tabata/amrap/emom), work_seconds, rest_seconds, rounds, current_round, is_running

7. EAT Module: Diet & Nutrition

Inspired by cult.fit Eat.fit and Freeletics nutrition coaching, this module provides a food database, daily meal logging, calorie tracking, weekly meal planning, and macro analysis fully integrated with the workout engine for net calorie calculations.

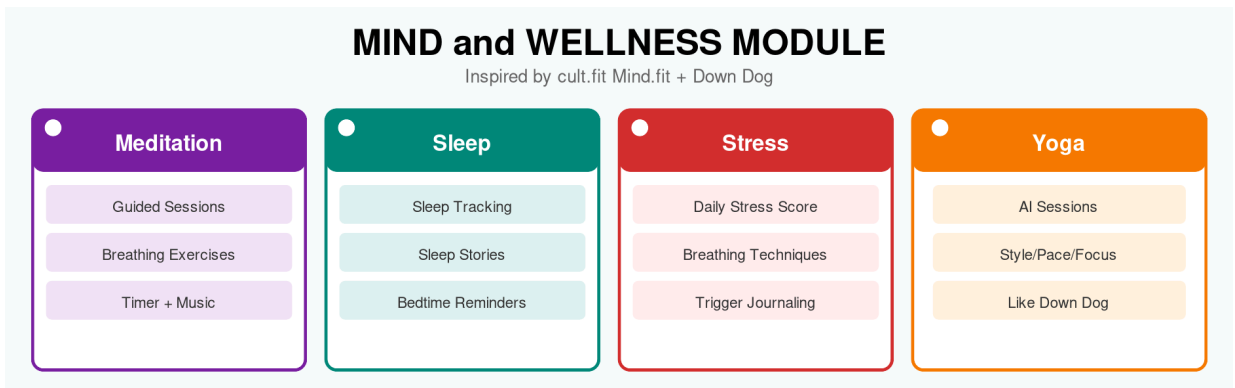


7.1 Models

Model	Key Fields
fitness.food.item	name, category (Selection: protein/carbs/dairy/vegetable/fruit/grain/snack/beverage), calories_per_serving, protein_g, carbs_g, fats_g, fiber_g, serving_size, serving_unit, barcode, is_verified, photo
fitness.meal.plan	member_id, name, week_start, goal_calories, meal_lines (O2M), is_template, created_by (trainer/ai/member)
fitness.meal.plan.line	plan_id, day (Selection: mon-sun), meal_type (breakfast/lunch/dinner/snack), food_item_id, quantity, total_calories (Computed)
fitness.meal.log	member_id, food_item_id, quantity, datetime, meal_type, total_calories (Computed), total_protein (Computed), photo
fitness.calorie.tracker	member_id, date, target_calories (from goal), consumed_calories (Computed from meal_logs), burned_calories (Computed from workouts), net_calories (Computed), protein_total, carbs_total, fats_total
fitness.water.intake	member_id, date, glasses_count, target_glasses, total_ml
fitness.recipe	name, ingredients (O2M food items), instructions (Html), prep_time, total_calories, diet_type (Selection: regular/vegan/keto/paleo), photo

8. MIND Module: Wellness & Meditation

Inspired by cult.fit Mind.fit's meditation and sleep features, Down Dog's AI-generated yoga sessions, and Fitbit's stress management, the MIND module brings holistic wellness into the fitness platform.



8.1 Models

Model	Key Fields
fitness.meditation	name, category (guided/breathing/sleep/focus/morning/evening), duration_minutes, audio_url, instructor, difficulty, description, play_count
fitness.meditation.log	member_id, meditation_id, datetime, duration_actual, mood_before, mood_after, notes
fitness.sleep.log	member_id, date, bedtime, wakeup_time, duration_hours (Computed), quality_score (1-10), deep_sleep_pct, notes
fitness.stress.log	member_id, date, stress_score (1-10), triggers (Text), coping_method, journal_entry (Text)
fitness.yoga.session	member_id, style (vinyasa/hatha/yin/power/restorative), focus_area (hips/back/core/full_body), duration, pace (slow/moderate/fast), difficulty, is_ai_generated, sequence_data (Json)

9. SOCIAL Module: Community & Gamification

Drawing from Strava's social network (14 billion kudos in 2025, 1M+ clubs), Peloton's Teams and High Fives, and cult.fit's energy meter gamification, the SOCIAL module transforms individual fitness into a connected community experience.



10. CARE Module: Tracking & Health Analytics

Inspired by Fitbit's comprehensive health metrics, Strava's performance analytics, and cult.fit's body composition analysis, the CARE module tracks physical health data and generates insights for both members and trainers.

10.1 Models

Model	Key Fields
<code>fitness.body.metric</code>	<code>member_id</code> , <code>date</code> , <code>weight</code> , <code>height</code> , <code>bmi</code> (Computed), <code>body_fat_pct</code> , <code>muscle_mass</code> , <code>waist</code> , <code>chest</code> , <code>hips</code> , <code>arms</code> , <code>thighs</code> , <code>photo_front</code> , <code>photo_side</code>
<code>fitness.activity.log</code>	<code>member_id</code> , <code>date</code> , <code>steps</code> , <code>distance_km</code> , <code>active_minutes</code> , <code>calories_burned</code> , <code>source</code> (manual/wearable), <code>heart_rate_avg</code> , <code>heart_rate_max</code>
<code>fitness.progress.snapshot</code>	<code>member_id</code> , <code>date</code> , <code>period</code> (weekly/monthly), <code>total_workouts</code> , <code>total_calories_burned</code> , <code>avg_workout_duration</code> , <code>streak_days</code> , <code>weight_change</code> , <code>strength_change_pct</code>
<code>fitness.health.report</code>	<code>member_id</code> , <code>date</code> , <code>report_type</code> (weekly/monthly/quarterly), <code>workout_summary</code> , <code>nutrition_summary</code> , <code>body_changes</code> , <code>recommendations</code> (Text), <code>pdf_attachment</code>
<code>fitness.readiness.score</code>	<code>member_id</code> , <code>date</code> , <code>score</code> (1-100), <code>sleep_component</code> , <code>activity_component</code> , <code>recovery_component</code> , <code>recommendation</code> (rest/light/moderate/intense)
<code>fitness.wearable.sync</code>	<code>member_id</code> , <code>provider</code> (fitbit/google_fit/apple_health/garmin), <code>last_sync</code> , <code>access_token</code> , <code>is_active</code>

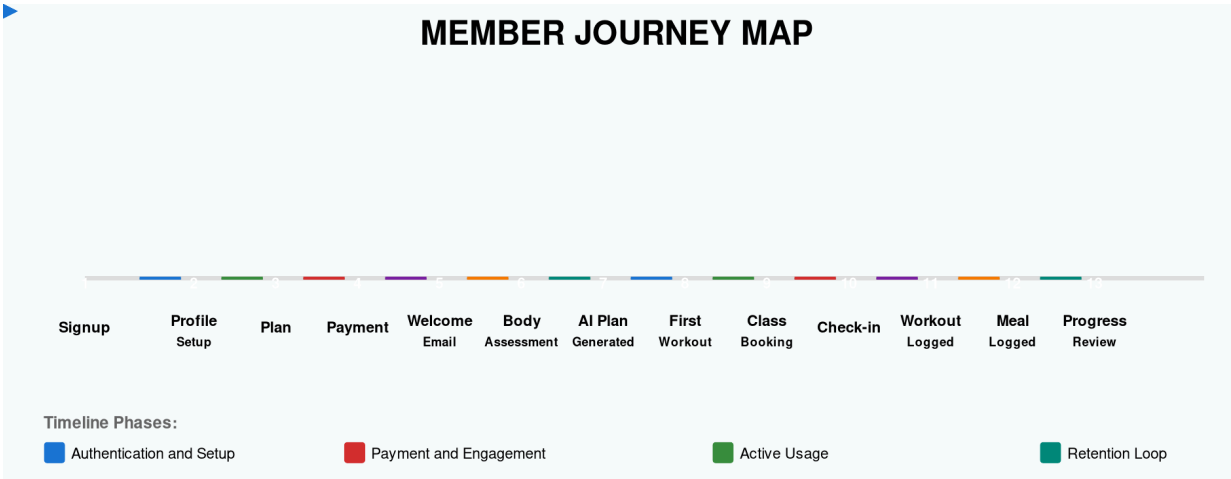
11. Business Layer: Subscriptions, Billing & CRM

The business layer manages all operational aspects: membership plans with cultpass-inspired tiers, subscription lifecycle, payment processing with Odoo account integration, equipment management, and multi-company support.

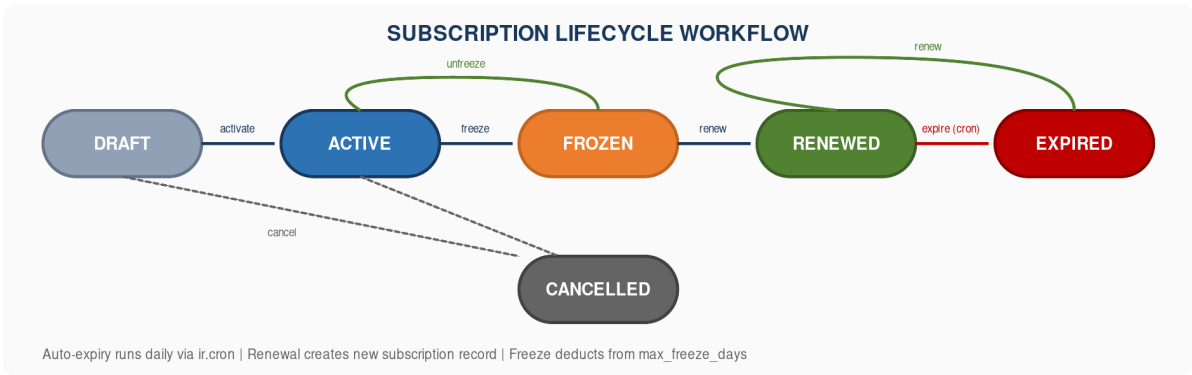
Model	Key Fields
fitness.membership.plan	name, plan_type (Selection: daily/weekly/monthly/quarterly/semi_annual/annual), tier (basic/pro/elite/platinum), price, features (Html), max_classes, max_freeze_days, includes_trainer, includes_nutrition, includes_wellness
fitness.subscription	name (Sequence), member_id, plan_id, start_date, end_date (Computed), state (draft/active/frozen/expired/cancelled), freeze_days_used, auto_renew, invoice_ids (M2M account.move)
fitness.payment	subscription_id, member_id, amount, payment_date, method (cash/card/bank/online), invoice_id (M2O account.move), state (pending/paid/failed/refunded), reference
fitness.equipment	name, category (cardio/strength/flexibility/accessories), brand, model, serial_number, purchase_date, cost, warranty_expiry, status (operational/maintenance/retired), location, photo, qr_code
fitness.equipment.maintenance	equipment_id, date, type (preventive/corrective/inspection), technician, cost, description, next_service_date
fitness.notification	member_id, type (workout_reminder/class_reminder/goal_alert/achievement/payment_due/subscription_expiry/challenge_update), title, message, is_read, sent_date, channel (app/email/sms)
fitness.reminder	member_id, type, time, days_of_week, is_active

12. Member Journey Map

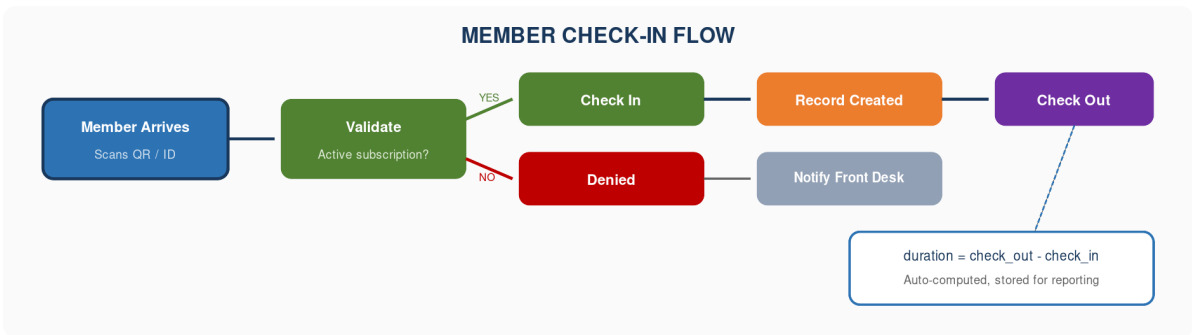
The complete member journey from first contact to long-term retention, showing how all six verticals connect:



12.1 Subscription Lifecycle

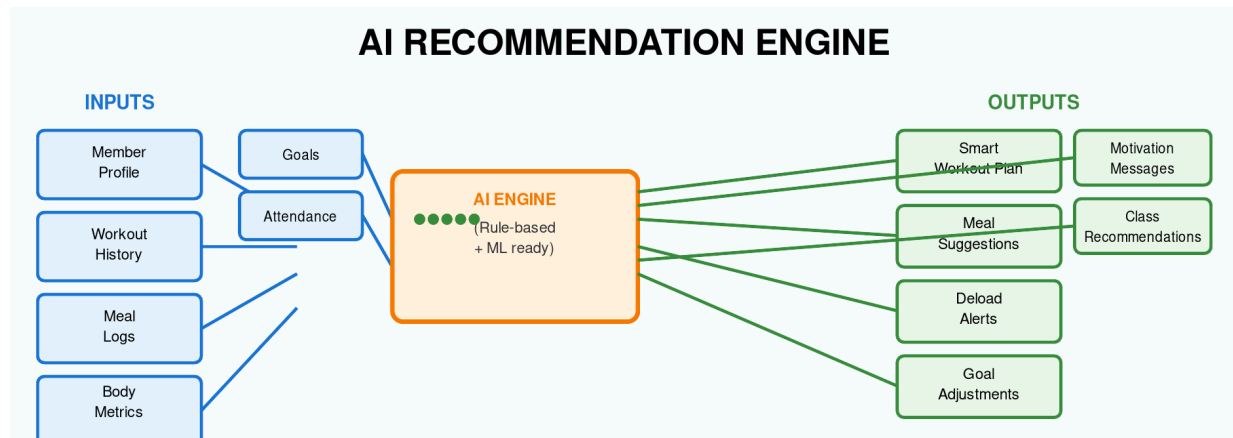


12.2 Check-In Flow



13. AI Recommendation Engine

Inspired by Freeletics AI Coach (trained on 56M+ users) and Fitbit's Gemini-powered health coach, our AI engine provides personalized recommendations across workouts, nutrition, and wellness. Phase 1 uses rule-based logic; Phase 2 upgrades to ML models.



13.1 AI Model

fitness.ai.recommendation: member_id, type (workout/meal/class/wellness/deload/goal_adjust), content (Text), reasoning (Text), confidence_score (Float), status (suggested/accepted/dismissed), generated_date, applied_date, source (rule_based/ml_model)

13.2 Rule-Based Logic (Phase 1)

- **Workout Selection:** Match exercises to member goal + available equipment + fitness level. Increase difficulty every 2 weeks if completion rate > 80%.
- **Deload Detection:** If 3+ consecutive sessions show declining performance (fewer reps or lower weight), suggest a deload week with 50% volume.
- **Meal Suggestions:** Based on daily calorie target minus consumed, suggest meals from database that fit remaining macro budget.
- **Class Recommendations:** Based on member preferences, past attendance, and schedule gaps, suggest upcoming classes with available spots.
- **Recovery Alerts:** If member worked same muscle group 2 days in a row, suggest rest or different focus area.
- **Streak Motivation:** If member is 1 day from breaking their longest streak, send motivational notification.

14. Complete Model Reference

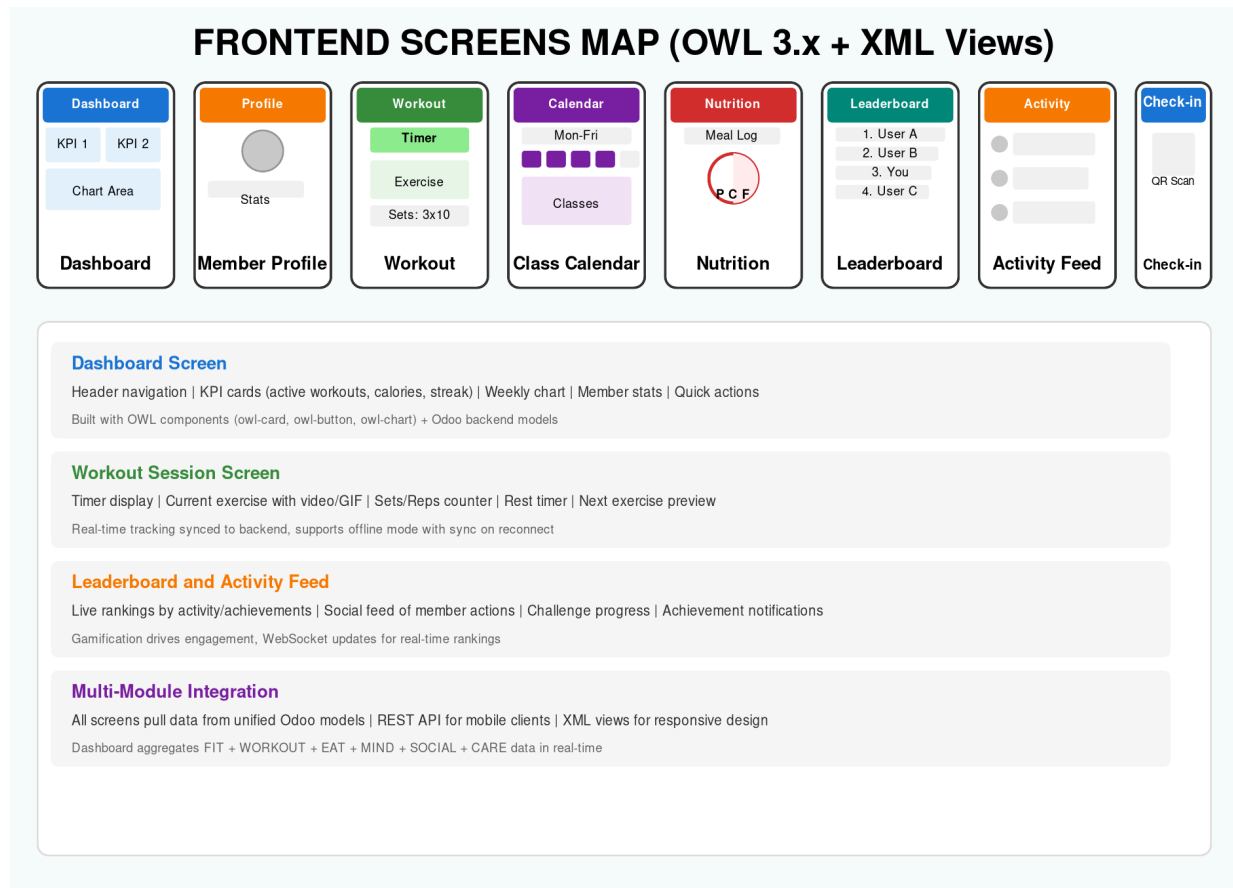
The platform consists of 50+ models across 7 groups. This is the complete reference:

Total Models: 49 | Verticals: 6 + Business Layer | OWL Components: 8

See sections 5-11 for detailed field specifications per model. The model naming convention is `fitness.{vertical}.{entity}`, ensuring clean namespace organization within Odoo.

15. OWL 3.x Frontend Components

All 8 OWL components are registered as `ir.actions.client` and accessible via menu items or embedded in form views. They use OWL 3.x patterns (required for Odoo 19) with Chart.js for visualizations.



16. Phased Development Plan

Instead of a rigid sprint, development is organized into four phases by product maturity. Each phase delivers a usable product increment:

PHASE 1: CORE PLATFORM: Foundation + Member + FIT + Basic Billing

- Module scaffold, security groups, base theme
- fitness.member + member.profile + member.goal models + views
- fitness.membership.plan + subscription + payment models + views
- fitness.class + schedule + booking + trainer models + views
- fitness.attendance + check-in wizard + kiosk OWL widget
- fitness.equipment + maintenance models + views
- Full menu hierarchy and navigation
- ACL + record rules for all Phase 1 models
- OWL Dashboard v1 (basic KPIs)
- QWeb PDF: member ID card, attendance report

PHASE 2: WORKOUT + NUTRITION: Exercise Engine + Meal Tracking + Timer

- fitness.workout.plan + plan.line + exercise + category models
- fitness.workout.session + log + timer models
- OWL Workout Timer widget (all modes: HIIT, Tabata, AMRAP, EMOM)
- Exercise library views with video preview and search
- fitness.food.item + meal.plan + meal.log + calorie.tracker
- fitness.water.intake + recipe models
- OWL Meal Tracker widget with macro charts
- Nutrition views: food search, meal planner, calorie dashboard
- Integration: calories burned from workouts into calorie tracker
- QWeb PDF: workout summary report, nutrition report

PHASE 3: SOCIAL + GAMIFICATION: Community Features + Leaderboards + Challenges

- fitness.friend + kudos + comment models
- fitness.activity.feed model + OWL Activity Feed widget
- fitness.leaderboard.entry model + OWL Leaderboard widget
- fitness.challenge + challenge.progress models + views
- fitness.achievement + member.achievement models
- fitness.club model + views
- Achievement trigger automation (cron)
- Leaderboard recalculation cron
- OWL Dashboard v2 (add social pulse panel)
- Notification system + email templates

PHASE 4: WELLNESS + AI + ANALYTICS: Mind Module + AI Engine + Health Reports

- fitness.meditation + meditation.log models
- fitness.sleep.log + stress.log models
- fitness.yoga.session model (AI-generated sequences)
- OWL Meditation Player widget
- fitness.body.metric + activity.log + progress.snapshot models
- OWL Body Scanner widget
- fitness.ai.recommendation model + rule-based engine
- fitness.health.report + readiness.score models
- fitness.wearable.sync model (API hooks)
- Full test suite + performance optimization + final packaging

17. Odoo 19 Technical Requirements

Mandatory development rules for Odoo 19:

- Python 3.12+ with type hints MANDATORY on all method signatures.
- SQL() builder for ALL raw SQL. No string concatenation or f-strings in queries.
- Views: <list> not <tree>. invisible/readonly use Python expressions directly (no attrs dict).
- Security groups: Use privilege_id NOT category_id for res.groups.
- Search views: group must use name=group_by only. Never expand or string attributes.
- Cron: No numbercall/doall attributes (removed). Use interval_number + interval_type.
- HTTP routes: type="jsonrpc" on all @http.route. type="json" REMOVED.
- res.partner: mobile field REMOVED. Store mobile directly on model.
- OWL 3.x required. Register components as ir.actions.client. Use type="action" for stat buttons.
- Manifest order: security XML -> ir.model.access.csv -> views -> menus -> data.
- Never check_company=True on res.company fields.
- Sale list views: Never ref sale.view_quotation_list (removed in v19).
- Module delivery: Always as single ZIP file in addons/ directory structure.

18. Website & Portal Layer

(Frontend-First)

A key differentiator of this module is the complete website and portal layer that allows members, trainers, and prospects to interact with the fitness center entirely through the Odoo website. This follows the frontend-first architecture: every backend model is designed to power a customer-facing experience. The portal extends Odoo's native portal framework (portal.mixin) while the public website uses website.published.mixin and QWeb templates for SEO-friendly, mobile-responsive pages.

18.1 Architecture Overview

The website/portal layer consists of three tiers:

- 1. Public Website Pages** — No login required. Visitors can browse membership plans, class schedules, trainer profiles, center locations, workout library, and recipes. Designed for SEO with website.seo.metadata mixin. 15 public pages across all phases.
- 2. Member Portal** — Authenticated pages under /my/fitness/. Members manage subscriptions, book classes, log workouts, track nutrition, participate in challenges, earn rewards, and view progress analytics. 19 portal pages.
- 3. Trainer Portal** — Authenticated pages under /my/trainer/. Trainers manage their class rosters, mark attendance, view client progress, and build custom workout templates. 4 trainer-specific pages.

18.2 Public Website Pages

Page	Route	Phase	Key Features	Models Used
Homepage / Landing	/	1	Hero banner, class carousel, trainer spotlight, pricing teaser	fitness.center, fitness.class.template
Membership Plans	/membership/plans	1	Plan cards, feature comparison, monthly/annual pricing	fitness.membership.plan
Plan Detail	/membership/plans/<id>	1	Full benefits, T&C, included classes, signup CTA	fitness.membership.plan
Membership Signup	/membership/signup/<id>	1	Multi-step form: info, health, plan, payment, promo code	fitness.member, fitness.subscription
Class Schedule	/classes	1	Calendar/list toggle, filter by category/trainer/center	fitness.class.schedule, fitness.class.template
Class Detail	/classes/<id>	1	Description, trainer, difficulty, upcoming schedule	fitness.class.template, fitness.trainer

Trainer Directory	/trainers	1	Cards with photo, specializations, ratings	fitness.trainer
Trainer Profile	/trainers/<id>	1	Bio, certifications, classes taught, reviews	fitness.trainer, fitness.feedback
Center Locations	/centers	1	Map view, location cards, amenities, hours	fitness.center
Center Detail	/centers/<id>	1	Areas, equipment, photos, class schedule	fitness.center, fitness.center.area
Workout Library	/workouts	2	Workout cards, filter by type/difficulty/muscle	fitness.workout.template
Recipe Library	/recipes	2	Recipe cards, diet tags, macro info, search	fitness.recipe
Challenges	/challenges	3	Active challenges, participant count, leaderboard	fitness.challenge
Blog / Articles	/blog	2	Fitness tips, nutrition articles, wellness content	blog.post
Contact / Inquiry	/contact	1	Contact form, center selector, inquiry type	crm.lead (optional)

18.3 Member Portal Pages (/my/fitness/)

All member portal pages extend Odoo's CustomerPortal class and render within portal.portal_layout. They provide full self-service capabilities so members rarely need to contact staff. Portal pages use access tokens (portal.mixin) for secure sharing and support both desktop and mobile layouts.

Page	Route	Phase	Key Features	Models Used
My Dashboard	/my/fitness	1	Subscription status, today's schedule, streak, stats	fitness.member, fitness.subscription
My Subscription	/my/fitness/subscription	1	Plan details, billing history, upgrade/downgrade	fitness.subscription, fitness.payment
My Bookings	/my/fitness/bookings	1	Upcoming/past classes, cancel, reschedule, waitlist	fitness.class.booking
Book a Class	/my/fitness/book/<id>	1	Booking flow, availability, waitlist, confirmation	fitness.class.booking, fitness.class.schedule
My Check-ins	/my/fitness/checkins	1	Visit history, frequency stats, streak display	fitness.checkin
My Profile & Health	/my/fitness/profile	1	Personal info, health data, goals, avatar	fitness.member, fitness.health.record

My Invoices	/my/fitness/invoices	1	Payment history, PDF download, status badges	fitness.payment, account.move
My Workouts	/my/fitness/workouts	2	Session history, exercise breakdown, personal records	fitness.workout.session
Start Workout	/my/fitness/workouts/start/<id>	2	Template preview, start, redirect to OWL player	fitness.workout.template
My Nutrition	/my/fitness/nutrition	2	Meal log, calorie ring, macro bars, water tracker	fitness.meal.entry, fitness.diet.goal
Log a Meal	/my/fitness/nutrition/log	2	Food search, barcode, portion, photo upload	fitness.meal.entry, fitness.food.item
My Meal Plans	/my/fitness/meal-plans	2	Weekly calendar, grocery list, nutritional summary	fitness.meal.plan
My PT Sessions	/my/fitness/pt-sessions	2	Appointments, book new, rate, trainer notes	fitness.pt.session
My Progress	/my/fitness/progress	2	Body measurements, trends, goal gauges, charts	fitness.body.measurement
My Challenges	/my/fitness/challenges	3	Active participation, progress, rankings	fitness.challenge.participant
My Badges & Rewards	/my/fitness/rewards	3	Badge gallery, points, tier, redemption catalog	fitness.reward.points, fitness.member.badge
Community Feed	/my/fitness/community	3	Posts, challenges, likes, comments, friend activity	fitness.community.post
My Wellness	/my/fitness/wellness	4	Meditation player, sleep log, mood journal	fitness.meditation, fitness.sleep.log
My Devices	/my/fitness/devices	4	Wearable connect/manage, sync settings	fitness.wearable

18.4 Trainer Portal (/my/trainer/)

Trainers have their own portal section with views tailored to their workflow: managing classes, tracking client progress, and building custom workout programs.

Page	Route	Phase	Key Features	Models Used
Trainer Dashboard	/my/trainer	2	Today's schedule, clients, class roster, earnings	fitness.trainer, fitness.class.schedule
My Classes	/my/trainer/classes	2	Class list, student roster, attendance marking	fitness.class.schedule, fitness.class.booking
My Clients	/my/trainer/clients	2	Client cards, plan overview, progress snapshot	fitness.pt.plan, fitness.member

Workout Builder	/my/trainer/workouts/create	2	Exercise picker, drag-drop ordering, save template	fitness.workout.template, fitness.exercise
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18.5 Technical Implementation

18.5.1 Module Dependencies

The website/portal layer requires the following Odoo depends in `__manifest__.py`: website, portal, payment, website_blog (optional). These are added to the existing fitness_center module depends list. The assets key in the manifest registers all frontend SCSS, JavaScript, and OWL components under web.assets_frontend.

18.5.2 Mixin Integration

Core models gain website and portal capabilities through mixin inheritance:

- **portal.mixin** added to: fitness.member, fitness.subscription, fitness.class.booking, fitness.workout.session, fitness.pt.session, fitness.payment — provides access_token and access_url
- **website.published.mixin** added to: fitness.membership.plan, fitness.class.template, fitness.trainer, fitness.workout.template, fitness.recipe, fitness.challenge — provides website_published and is_published toggle
- **website.seo.metadata** added to all website.published models — provides website_meta_title, website_meta_description, og:image for SEO optimization

18.5.3 Controller Architecture

Three controller classes handle all routes:

- **FitnessWebsiteController(http.Controller)** — 16 public routes for the website pages. Uses auth="public", website=True. All queries use sudo() for public access. Includes AJAX endpoints (type="jsonrpc") for dynamic filtering on class schedule and food search.
- **FitnessMemberPortal(CustomerPortal)** — 24 routes for member self-service. Extends Odoo's CustomerPortal with _prepare_home_portal_values for counters. Uses auth="user". Includes 4 AJAX endpoints for booking, meal logging, community posts, and reward redemption.
- **FitnessTrainerPortal(CustomerPortal)** — 5 routes for trainer portal. Verifies trainer role via record rules. Includes AJAX endpoint for attendance marking.

18.5.4 QWeb Templates

The module includes 40 QWeb templates organized as: 11 public website page templates (inheriting website.layout), 20 portal templates (inheriting portal.portal_layout), 5 trainer portal templates, and 4 website builder snippets (membership pricing table, class carousel, trainer spotlight, testimonial carousel). All templates use Bootstrap 5 classes for responsive design and follow Odoo's website builder compatibility patterns.

18.5.5 Portal Security

Portal access is secured through ir.rule record rules for base.group_portal:

- **Members** can only read their own records: subscriptions, bookings, workouts, nutrition, challenges, rewards, invoices (domain: partner_id = user.partner_id.id)
- **Trainers** can read their assigned classes, clients, and PT sessions (domain: trainer_id.partner_id = user.partner_id.id)
- **Write access** is granted only for specific actions: booking creation, meal logging, profile editing, community posting, and workout session recording

18.5.6 Frontend Assets

Custom frontend assets are registered in the manifest under web.assets_frontend:

- **SCSS** — Custom styles for fitness pages, dark/light theme support, responsive breakpoints, animation classes for progress bars and charts
- **JavaScript** — Public widgets for dynamic filtering (class schedule, food search), AJAX form submissions, interactive charts (Chart.js), barcode scanner integration, and the OWL 3.x components that render within portal pages
- **PWA (Phase 4)** — Progressive Web App configuration with service worker for offline support, manifest.json for home screen installation, and push notification subscription

18.6 Key User Journeys (Website → Portal)

Journey 1: New Member Signup

Visitor lands on homepage (/) → browses membership plans (/membership/plans) → views plan detail → clicks "Join Now" → fills multi-step signup form (/membership/signup/<id>) with personal info, health questionnaire, applies promo code → completes payment → auto-creates fitness.member + fitness.subscription + portal user → redirected to member dashboard (/my/fitness) → sees first-time onboarding tutorial.

Journey 2: Class Booking

Member logs into portal → views class schedule (/classes or /my/fitness/bookings) → filters by yoga category → finds available slot → clicks "Book" → system creates fitness.class.booking → confirmation email sent → class appears in "My Bookings" with calendar export link → day of class: member checks in via kiosk → trainer marks attendance.

Journey 3: Workout + Nutrition Tracking

Member visits /my/fitness/workouts → browses workout library → selects "Upper Body Strength" → clicks "Start Workout" → redirected to OWL WorkoutPlayer component → completes workout with set/rep logging → session saved → navigates to /my/fitness/nutrition → logs post-workout meal with barcode scan → calorie ring updates → views progress at /my/fitness/progress.

Journey 4: Trainer Workflow

Trainer logs into /my/trainer → sees today's 3 group classes and 2 PT sessions → clicks into 9am Yoga class → views roster of 18 enrolled members → marks attendance as members check in → adds session notes → navigates to PT client → reviews client's workout history and body measurements → creates custom workout template using drag-drop builder → assigns to client's plan.